

CYBERSECURITY PROJECT REPORT

Home Network Security Assessment Using Kali Linux and Nmap

A Practical Network Security Assessment Using Kali Linux and Nmap

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1. Project Overview

1.1 Introduction

Network security assessments are an essential part of cybersecurity, helping to identify exposed services, potential security risks, and system misconfigurations. By examining a system's network exposure, security professionals can evaluate whether services are configured securely and recommend improvements to reduce the attack surface. This project focuses on assessing a Windows host within a controlled home laboratory environment using Kali Linux and Nmap.

1.2 Purpose of the Project

The purpose of this project is to perform a basic home network security assessment by identifying active network services, analyzing their security implications, and documenting recommendations for improving system security. The project also aims to develop practical skills in network reconnaissance, service enumeration, and security analysis using industry-standard tools.

1.3 Importance of Network Security Assessments

Network security assessments help identify open ports, active services, and potential security weaknesses before they can be exploited. Regular assessments improve system security, reduce the attack surface, verify that only necessary services are exposed, and support proactive risk management.

1.4 Project Goals

The primary goal of this project was to build a controlled cybersecurity lab using Oracle VirtualBox and Kali Linux, perform a network security assessment on a personally owned Windows computer, analyze the discovered network services, and document the findings in a professional security assessment report. Through this project, practical experience was gained in network reconnaissance, service analysis, risk assessment, and technical documentation.

2.Objective

2.1 Primary Objective

To perform a home network security assessment on a Windows host using Kali Linux and Nmap to identify active network services, evaluate potential security risks, and recommend appropriate security improvements.

2.2 Secondary Objective

- Build a controlled cybersecurity lab using Oracle VirtualBox and Kali Linux.
- Identify the target system's IP address and verify network connectivity.
- Perform network reconnaissance and port scanning using Nmap.
- Enumerate active network services and analyze their functions.
- Investigate running network processes using Windows command-line tools.
- Assess the security implications of identified services.
- Document findings, risk assessments, and recommendations in a professional security report.

3.Scope

3.1 In Scope

This project focused on assessing the network exposure of a personally owned Windows 11 computer within a controlled home laboratory environment. The assessment included network reconnaissance, connectivity testing, port scanning, service enumeration, and basic process investigation using Kali Linux and Windows command-line tools. Only authorized systems owned by the user were included in the assessment.

3.2 Out of Scope

The assessment did not include vulnerability exploitation, password attacks, malware analysis, denial-of-service testing, wireless network testing, or scanning of third-party or public systems. Internet-facing services and cloud-based resources were also excluded.

3.3 Limitations

This project was limited to identifying active ports and services and performing a basic security analysis. It did not include advanced penetration testing, exploit validation, or comprehensive vulnerability scanning. The findings are based on the system configuration at the time of the assessment.

4.Lab Setup

4.1 Overview

A controlled home laboratory environment was created to perform the network security assessment safely and ethically. The lab consisted of a Windows 11 host system and a Kali Linux virtual machine running on Oracle VirtualBox. Both systems were connected to the same home network to enable communication and security testing.

4.2 Hardware Configuration

Component	Specification
Host Computer	Windows Laptop/Desktop
Processor	11 th Gen Intel(R) Core (TM) i5-1135G7@2.40GHz
Memory (RAM)	8 GB
Storage	153 GB

4.3 Software Configuration

Software	Version	Purpose
Windows 11	Current	Target Operating System
Oracle VirtualBox	<i>Version 7.2.6 r172322</i>	Virtualization Platform
Kali Linux	<i>Version 2026.1</i>	Security Assessment Machine
Nmap	<i>Version 7.98</i>	Port Scanning & Service Enumeration

4.4 Network Configuration

Component	Details
Network Type	Home Network
Target System	Windows 11
Assessment System	Kali Linux Virtual Machine
Communication	Local Area Network (LAN)

4.5 Test Environment

The assessment was conducted on a personally owned Windows computer within a controlled home network. Kali Linux was deployed as a virtual machine using Oracle VirtualBox to perform authorized network reconnaissance and service analysis. All testing activities were limited to systems owned by the user, ensuring compliance with ethical cybersecurity practices.

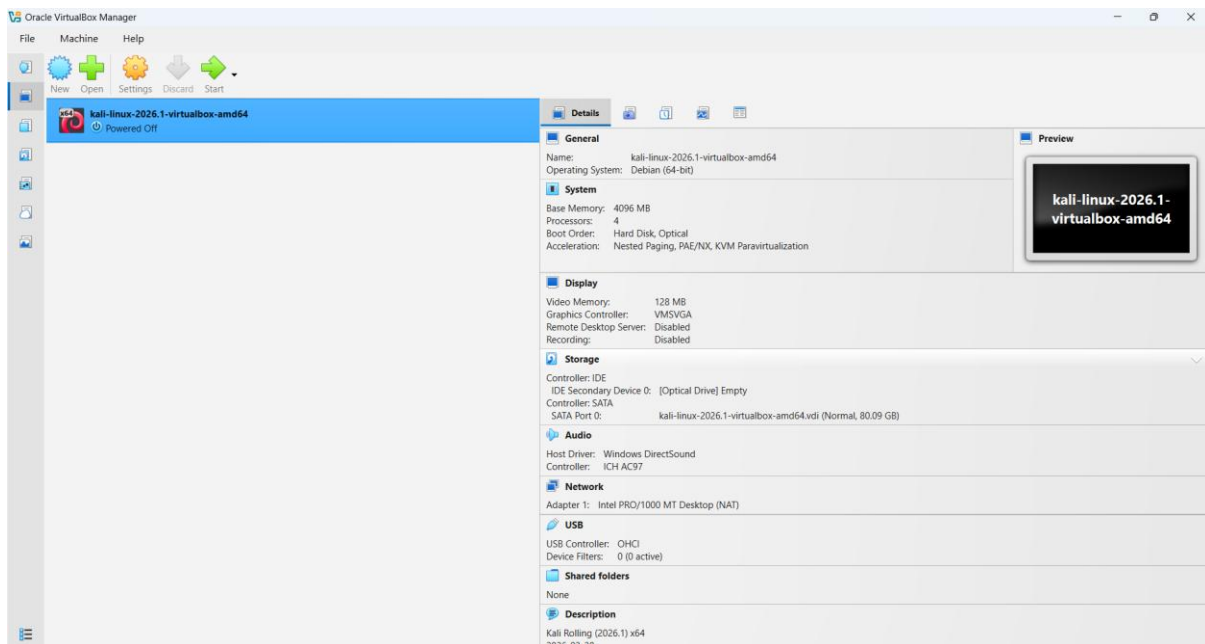


Figure 1: Oracle VirtualBox Manager

4.6 Lab Architecture

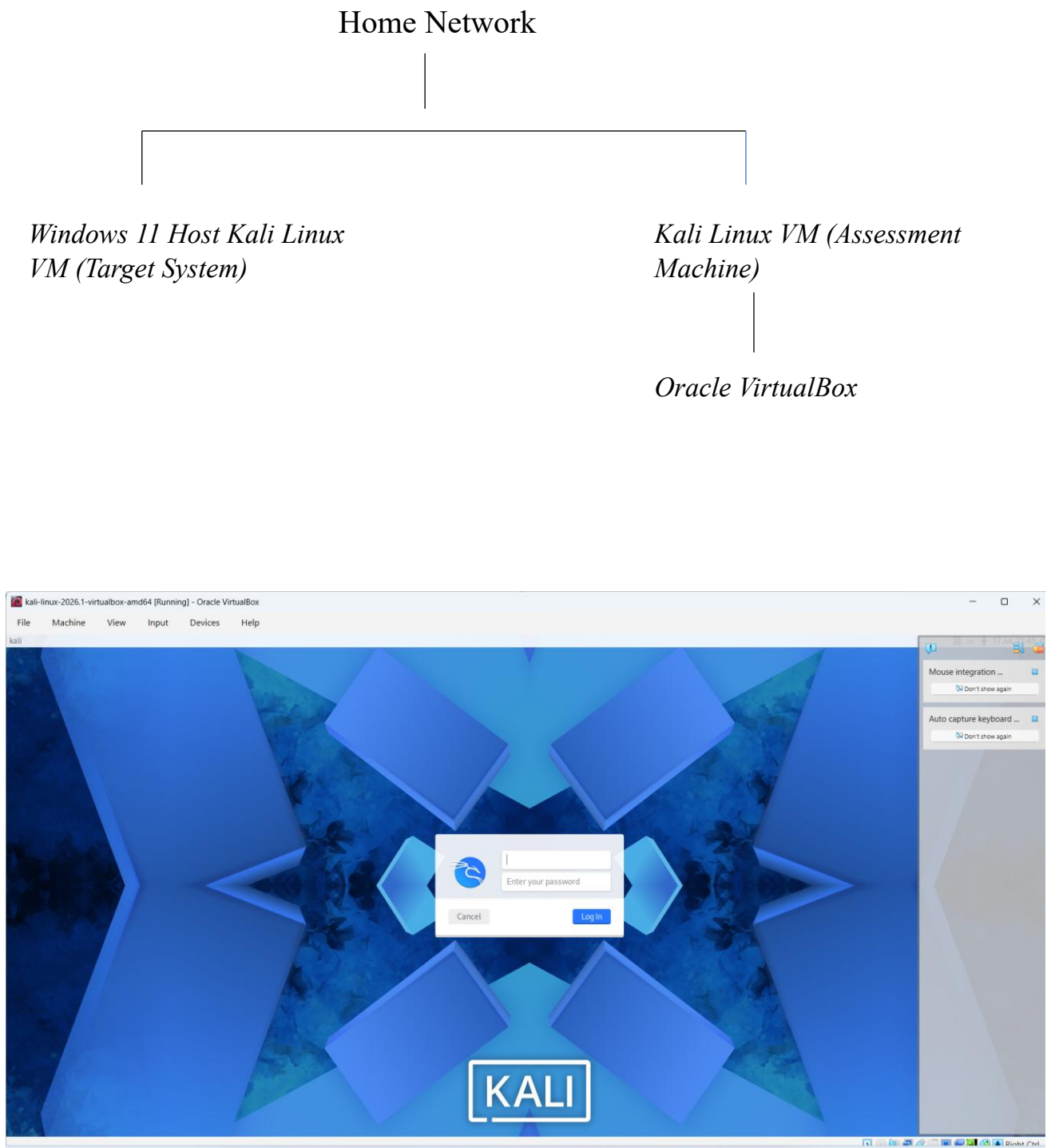


Figure 2: Kali Linux Virtual Machine

5.Tools Used

Description

Several industry-standard tools were used throughout this project to conduct the network security assessment. Oracle VirtualBox hosted the Kali Linux virtual machine, while Kali Linux provided the environment for performing network reconnaissance and analysis. Nmap was used to identify active hosts, open ports, and network services. Windows Command Prompt utilities, including `ipconfig`, `ping`, `netstat`, and `tasklist`, were used to gather system and network information, verify connectivity, and investigate running processes associated with the identified services. Together, these tools enabled a structured and authorized assessment of the home network environment.

The following tools were used to perform the home network security assessment:

Tool	Purpose
Oracle VirtualBox	Used to create and run the Kali Linux virtual machine for conducting the security assessment.
Kali Linux	Security-focused Linux distribution used as the assessment platform.
Nmap (Network Mapper)	Used to discover live hosts, identify open ports, and enumerate running network services.
Windows Command Prompt (CMD)	Used to execute commands such as <code>ipconfig</code> , <code>ping</code> , <code>netstat</code> , and <code>tasklist</code> for network and system information.
ipconfig	Displayed the IP configuration of the Windows host.
ping	Verified network connectivity between the Kali Linux virtual machine and the Windows host.
netstat	Displayed active network connections, listening ports, and associated services.
tasklist	Listed running processes on the Windows system to assist in service identification.

6.Methodology

. The methodology consisted of the following phases:

6.1 Lab Preparation

A Kali Linux virtual machine was created using Oracle VirtualBox and connected to the same home network as the Windows 11 host. This provided a controlled environment for conducting the security assessment.

6.2 Network Identification

The IP address of the Windows host was identified using the `ipconfig` command. Network connectivity between the Kali Linux virtual machine and the Windows host was verified using the `ping` command to ensure successful communication.

6.3 Network Scanning

Nmap was used to scan the target system and identify open ports and active network services. A standard port scan was performed to discover listening ports, followed by service version detection to gather additional information about the services running on the target.

6.4 Service Analysis

The identified network services were analyzed to understand their functions and assess any potential security implications. Active connections and listening ports were reviewed using the `netstat` command, while the `tasklist` command was used to examine running processes associated with the discovered services.

6.5 Risk Assessment

Each identified service was evaluated based on its purpose, potential security risks, and exposure. Risk levels were assigned according to the likelihood of misuse if the service were left unsecured or improperly configured.

6.6 Recommendations

Based on the assessment findings, practical security recommendations were developed to reduce the attack surface, improve system security, and encourage the use of security best practices.

7. Commands Used

The following commands were executed during the home network security assessment to collect network information, verify connectivity, identify active services, and investigate running processes. Each command contributed to the overall analysis of the target system.

Table 5. Commands Used During the Assessment

Command	Purpose	Outcome
<code>ipconfig</code>	Display the IP configuration of the Windows host.	Identified the target system's IP address.
<code>ping <IP Address></code>	Verify network connectivity between Kali Linux and the Windows host.	Confirmed successful communication between both systems.
<code>nmap <IP Address></code>	Perform a basic port scan to identify open ports.	Discovered open TCP ports on the target system.
<code>nmap -sV <IP Address></code>	Detect service versions running on open ports.	Identified services associated with the discovered ports.
<code>netstat -ano</code>	Display active network connections and listening ports.	Verified active ports and their associated process IDs (PIDs).
<code>tasklist</code>	List all running processes on the Windows system.	Helped identify processes related to the discovered services.

8.Findings

The network security assessment successfully identified several active network services on the Windows 11 host. The Nmap scan revealed multiple open TCP ports associated with standard Windows services and one additional service running on port **12345**. The results were verified using Windows command-line utilities, including `netstat -ano` and `tasklist`, to confirm active connections and identify the corresponding running processes.

The assessment identified the following open ports and services:

Table 6. Network Assessment Findings

Port	Service	Description	Risk Level
135/TCP	Microsoft Windows RPC	Supports communication between Windows services and applications.	Medium
139/TCP	NetBIOS Session Service	Used for legacy file and printer sharing over a local network.	Medium
445/TCP	Microsoft SMB	Provides file and printer sharing services on Windows systems.	Medium–High
12345/TCP	ElevationService.exe	A locally running service identified during the assessment that should be verified as a legitimate application.	Low–Medium

9. Risk Assessment

The identified network services were assessed based on their purpose, potential security impact, and likelihood of misuse if exposed or improperly configured. The assessment focused on identifying possible risks associated with the discovered services rather than confirming the existence of vulnerabilities. Risk levels were assigned using a qualitative approach (Low, Medium, and High) based on the potential effect on system security.

Table 7. Risk Assessment

Port	Service	Risk Level	Justification
135/TCP	Microsoft Windows RPC	Medium	Required for Windows communication but may become a target if unnecessarily exposed to untrusted networks.
139/TCP	NetBIOS Session Service	Medium	Used for legacy file sharing and should be disabled if not required to reduce unnecessary exposure.
445/TCP	Microsoft SMB	Medium–High	Supports file and printer sharing. If misconfigured or left exposed, it can increase the attack surface.
12345/TCP	ElevationService.exe	Low–Medium	An active service that should be verified as a legitimate application and monitored to ensure it is functioning as intended.

10.Recommendations

Based on the findings of the network security assessment, several recommendations are proposed to improve the overall security of the Windows host. These recommendations aim to reduce the system's attack surface, strengthen network security, and promote good cybersecurity practices.

- Keep the Windows operating system and installed software up to date by applying the latest security patches and updates.
- Enable and properly configure the Windows Firewall to restrict unnecessary inbound and outbound network traffic.
- Disable network services such as NetBIOS or SMB if they are not required for normal system operation.
- Regularly review open ports and running services to identify and remove unnecessary or unused applications.
- Verify that services such as **ElevationService.exe** belong to trusted and legitimate software and remove any applications that are no longer needed.
- Use strong passwords and enable multi-factor authentication (MFA) wherever possible to enhance account security.
- Install and maintain reputable antivirus or endpoint protection software to detect and prevent malicious activity.
- Perform periodic network security assessments to identify configuration changes, newly exposed services, and potential security risks.

11. Conclusion

This project successfully demonstrated the process of conducting a basic home network security assessment using Kali Linux and Nmap within a controlled and authorized laboratory environment. The assessment identified active network services, verified network connectivity, and analyzed the security implications of the discovered ports using standard cybersecurity tools and Windows command-line utilities.

The findings showed that the Windows host was running several common services required for normal system functionality. Based on these findings, practical security recommendations were provided to help reduce the system's attack surface and improve its overall security posture. No vulnerability exploitation or unauthorized testing was performed during the assessment, ensuring that the project adhered to ethical cybersecurity practices.

Overall, this project provided valuable hands-on experience in network reconnaissance, service enumeration, risk assessment, and technical documentation. It also strengthened practical skills in using industry-standard cybersecurity tools and reinforced the importance of conducting regular network security assessments to maintain a secure computing environment.